

CETM25: Assignment 2

Development and Project Management Report

A Report of the Design, Development and Management of a Big Data Visualisation Product Project

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Introduction

This report critically evaluates the development and project management of a proof-of-concept Big Data visualisation product built using Python and the Streamlit framework. The product presents Universities and Colleges Admissions Service (UCAS) undergraduate applicant data (UCAS, 2026) across three academic years (2023, 2024 and 2025) to enable junior government officials to deliver data storytelling to professional data-driven journalists. The report builds directly on the design plan submitted in assignment one and reflects on how initial intentions were refined, extended or revised during development. Knaflic (2015) argues that effective data communication requires not only technical expertise but also deliberate design thinking that is centred on the needs of the audience. This principle has informed the decisions made throughout this project.

Section 1. Development

1.1 Selection of Visualisation Software and Hardware

The software architecture selected in assignment one was retained in assignment two development. Python was chosen as the primary programming language, with the Streamlit framework providing the interactive web application layer. Streamlit enables rapid, iterative deployment of browser-accessible data applications without requiring front-end web development expertise, making it particularly well-suited to this project's requirement for non-technical user deployment (Saraswat, 2023). As it is hosted on the Streamlit Community Cloud (Toolify.ai, 2024) it can be accessed via a public URL which satisfies the functional requirement that no additional software installation is needed.

The prototype dashboard is available to view at <https://mjhalpin.streamlit.app>

The core coding libraries used were NumPy and Pandas for data ingestion, cleaning, reshaping and aggregation, and Plotly Express for interactive chart rendering. Plotly was selected over native Streamlit charts as it provides greater control over visual functionality. Plotly offers hover over tooltips, grouped bar chart orientation and control of legend positioning which are essential for delivering a professional, accessible dashboard. Code development used a local Visual Studio Code (VS Code) integrated development environment (IDE) with a Python virtual environment to manage package dependencies and ensure reproducibility. VS Code was selected for its extensive Python support and version control integration via GitHub. The application was deployed to Streamlit Community Cloud from the GitHub repository which enables continuous deployment on each code commit.

1.2 Project Development / Software Engineering Methodology

Project development adopted an Agile, iterative methodology. Rather than specifying and building all features before testing, the project progressed through repeated cycles of build, test and refine spread evenly across several weeks. Suetin et al. (2016) argue

that flexible Agile methodologies outperform traditional waterfall approaches in contexts where implementation requirements are not fully established at the start of a project. This very much reflected the iterative development needed to determine optimal chart type, filter architecture and layout hierarchy that could be achieved using the Streamlit tool. This approach aligns with the Shore et al. (2021) description of Agile development as a discipline of continuous feedback rather than a fixed linear progression.

Data preprocessing was handled within the Python script rather than manually in the source CSV. Two functions were written: `clean_data()` and `simpler_cats()`. The `clean_data()` function performs column name normalisation, type coercion, removal of the aggregate 'All' rows, title case standardisation and null value removal. The `simpler_cats()` function creates three new categorical columns using data derived from existing variables:

- 'Maturity': values are Traditional or Mature
- 'Male/Female': values are Male, Female or Other
- 'Location': values are UK, EU, Not-EU)

Creation of these categorical aggregations was necessary to enable granular and simplified dashboard filter functionality without modifying the raw source file. Both functions include try/except error handling with descriptive error messages should future data ingest files contain incompatible content. To satisfy the assignment requirement for three comparable annual time points, the dataset was filtered to exclude years prior to 2023.

One significant design departure from the assignment one plan was the decision not to implement the planned choropleth map visualisation. During development, a working UK choropleth was successfully produced using GeoPandas and ONS GeoJSON boundary files. However, several practical problems emerged:

- Domicile data grouped non-UK applicants only as 'EU' or 'Non-EU' rather than by country, preventing 'like-for-like' geographic comparisons of non-UK countries.
- The map was difficult to render effectively on mobile screen sizes.
- Sequential colour schemes reflecting applicant volumes risked visual misinterpretation and failure to meet visual impairment (Colour Blind Awareness, 2026) accessibility requirements (W3C (2025)) .

As Strachnyi (2022) notes, colour selection in data visualisation carries significant interpretative risk particularly when sequential palettes are applied to data that is not inherently ordered. Therefore, the decision was made to replace the choropleth map with interactive line and bar charts that are arguably a better fit for a dynamic Streamlit layout and a public user audience that has varying digital literacy levels audience (Wlekly, 2024).

1.3 System Testing

System testing that is consistent with the exploratory testing principles described by Sommerville (2016) was conducted throughout development; instead of a discrete final phase, testing was iteratively performed as each component was developed. The following test categories were applied:

1.3.1 Data integrity testing

The `clean_data()` function was verified against the source CSV to confirm that column names were correctly normalised, numeric coercion was applied to the Applicants and Year columns and that rows containing aggregate totals ('All') were successfully excluded. Results were inspected using Pandas `.head()`, `.dtypes` and `.value_counts()` methods during development.

1.3.2 Filter interaction testing

All combinations of the primary legend selector (Age Group, Gender, Domicile) and associated sub-filters were tested to confirm that the KPI metric cards, line chart and bar chart updated correctly and that no combination produced an application error.

1.3.3 Edge case testing

The behaviour of the dashboard when all sub-filter values were deselected was tested.

1.3.4 Cross-device testing

The dashboard was tested on both desktop and mobile browsers. Mobile testing showed that users were not able to access the chart filter functionality. This error occurred as an unintended consequence of steps taken to hide developer icons from the public access interface. This was resolved by amending the Python script custom CSS to ensure that only the developer elements were hidden from the user interface.

1.3.5 Export testing

The CSV download button was tested to confirm that the exported file correctly reflected the currently filtered view and was labelled with the active legend selection.

1.3.6 Data refresh

This report advocates Google Cloud Storage (GCS) as the preferred secure and ease-of-use data refresh solution for Streamlit apps (Streamlit, 2026). However, as purchase of a GCS subscription is beyond the scope of this prototype proof-of-concept development, it was not possible to test data refresh via ingest of a new csv file.

1.4 User Evaluation

Although formal user evaluation was not conducted as part of this prototype development phase, a structured evaluation plan has been designed. As Munzner (2025) emphasises, visualisation systems should be validated against actual user requirements and digital literacy levels rather than assumed preferences. Given the two distinct dashboard user groups of junior government officials (operators) and data journalists (viewers), separate evaluation approaches are advised.

For junior government officials, a task-based usability test would be the most appropriate evaluation method. Participants could be asked to complete defined tasks such as “Filter the chart to show only female applicants from 2023 to 2025”. Performance could then be measured using a standardised instrument such as the System Usability Scale (SUS) (Brooke, 1996) which would provide a quantifiable usability score comparable across iterations of the product.

For data journalists, a qualitative evaluation using semi-structured interviews is arguably a more appropriate approach; with a focus on whether the dashboard surfaced meaningful data and whether the data story was compelling and trustworthy. Questions could include: “What trend surprised you most?”, “Did the visualisation help you identify a story?” and “Would you trust this data to support a published article?”. Mapping responses back to the functional and non-functional requirements defined in Assignment One could help confirm whether the product met its original specification to communicate the following key facts:

- Year-on-year applicant volume by domicile.
- Proportional shift in EU vs non-EU international applicants post-2021.
- Gender distribution trends over the full period.
- Changing ratio of first-time applicants to reapplicants.

Section 2. Project Management

2.1 Time Management

The project workload was distributed evenly across three weeks from the submission of assignment 1 through to the assignment 2 deadline. This even distribution was deliberate: having established the design plan in assignment one, development could proceed incrementally enabling iterative testing development:

Week 1: establish development environment and confirm data pipeline.

Week 2: iterative feature development and testing.

Week 3: documentation, report writing and final review.

The main time management challenge arose during the choropleth map exploration phase which consumed approximately two to three days before a decision was taken to pivot to an alternative approach. Kerzner (2022) advocates critical path analysis as a technique for identifying task dependencies that carry the greatest schedule risk. In this project, the initial selection of choropleth chart and requirement to change this choice, represents a high-dependency risk. In retrospect, applying a formal time-boxing limit (Miranda, 2011) on exploratory work would have triggered an earlier decision to abandon use of the choropleth map. Future projects would benefit from explicit time limited durations for research and high-risk activities.

The parallel demands of postgraduate study alongside professional commitments also presented an ongoing time pressure. Structuring the project as a series of small, achievable weekly milestones rather than large sequential phases, helped maintain momentum and ensured that no single issue could interrupt overall progress.

2.2 Risk Assessment

Risk identification was conducted systematically using a qualitative probability and impact matrix approach, consistent with best practice in project management methodology (Wingate, 2025). Each identified risk was scored on a Likert scale of 1 to 5 for both likelihood of occurrence and potential impact. The combined score (likelihood × impact) is used to classify risks as high (≥15), medium (9–14) or low (<9) as show in table 1.

Table 1. Project Risk Assessment

Risk	Likelihood	Impact	Score	Mitigation
Data schema change UCAS changes CSV column structure, breaking preprocessing pipeline.	4	5	20	README file guidance explains required structure of CSV files.
Visual misinterpretation Users unable to interpret charts.	4	4	16	Include dashboard headings, subheadings, chart annotations and narrative text that contextualises visuals.
Data refresh complexity Non-technical junior government operators unable to update source CSV securely.	3	4	12	Use Google Cloud Storage identified as secure solution with README markdown file guidance.
Platform dependency Streamlit Community Cloud discontinued or paywalled.	2	3	6	Application is containerisable to enable easy migration to an alternative platform.
Scope creep Feature additions extending beyond proof-of-concept brief.	2	3	6	Apply Agile time-boxing in subsequent development.
Accessibility non-compliance Colour choices inaccessible to visually impaired users.	1	3	3	WCAG 2.1 compliant colour palettes and includes alternative accessible data format.
Mobile rendering issues Dashboard layout breaks on small screens.	1	3	3	Mobile browser testing, refined CSS to preserve sidebar toggle, Streamlit responsive stacking confirmed.

The highest-scoring risks were data schema change and visual misinterpretation, both of which were actively mitigated during development. The data refresh risk represents a known residual risk for any production deployment. As Wingate (2025) notes, the purpose of a risk register is not to eliminate uncertainty but to make it visible and manageable. Product implementation and would explicitly highlight this within customer communications.

2.3 Quality Control

Project quality control has operated at three levels: code quality, data quality, and visualisation quality. Code quality was maintained through use of clear docstrings, consistent naming conventions and inline comments explaining sequence of code actions. The use of a Python virtual environment and a requirements.txt file ensures that the application can be reproduced consistently across different machines.

Data quality was assured by sourcing data exclusively from UCAS, a nationally recognised, publicly accountable body and by applying data cleaning steps that are transparent, repeatable and version-controlled via GitHub. This approach supports the principle of reproducible research, in which data transformations are documented and auditable (McKinney, 2022).

Visualisation quality was guided by the dashboard design principles articulated by Few (2013), particularly the recommendation to prioritise clarity and structure over decorative elements. The dashboard layout was designed to engage and guide users with varying data literacy levels through the dashboard; aggregated KPI metrics are presented first followed by trend visualisations and concluding with a granular data table. All charts were reviewed for correct axis labelling, legend clarity and appropriate chart type selection for the intended audience.

2.4 Customer Relationship Management

In this project, the customer relationship exists between the development team and the commissioning government team client responsible for deploying the product. Effective customer relationship management in software development requires clear communication, managed expectations and regular delivery of tangible output (Kinnett, 2017). The Assignment One report functioned as a formal requirements document, establishing a shared understanding of scope, data source, audience and technical approach prior to development. In a live client engagement, this would be supplemented by regular sprint reviews to demonstrate working functionality at the end of each development iteration. This practice aligns with Agile principles (Woodward et al., 2010) that aim to efficiently identify any misalignment between product and client expectations before it becomes costly to correct.

The user README file submitted alongside this report aims to offer an accessible guide for non-technical operators, enabling them to deploy and use the product independently and reduce ongoing development team support.

2.5 Product Market Strategy

The core value proposition of the proof-of-concept UCAS dashboard is the democratisation of complex data; the transformation of structured public interest datasets into accessible, interactive narratives that non-specialist audiences can engage with and act upon. This aligns with a growing demand across public bodies for self-service business intelligence tools that reduce dependence on specialist data science teams (Taei, 2017).

The start-up company go-to-market strategy could pursue two parallel routes. The first is a government business-to-business (B2B) route (Seebacher, 2021), positioning the product as a low-cost, rapidly deployable alternative to enterprise business intelligence platforms such as Tableau or Microsoft Power BI. Streamlit-based solutions offer a significantly lower

total cost option narrow-use-case dashboards; a full business intelligence platform licence would outweigh project deployment requirements. The second is a media partnership model in which dashboards are co-branded with public media organisations and used to drive engagement with data journalism content, increasing both public reach and the commissioning organisation's profile (Varroney, 2022).

The UCAS dataset itself is a compelling use case example: higher education access is currently a topic of sustained public and political interest and the dashboards capacity to highlight intersectional trends such as the post-Brexit decline in EU applicants or the changing ratio of mature to traditional-age students can generate data stories that drive media coverage. As Segel and Heer (2010) argue, narrative visualisation that combines an authored story with open-ended exploration is particularly effective at sustaining audience engagement. Used as a portfolio piece or live demonstration, this prototype could credibly support business development conversations with government departments, higher education institutions or public media outlets.

3. Conclusion

This report has critically evaluated the development and project management of a Streamlit-based UCAS applicant data visualisation dashboard. An iterative, Agile development approach has enabled responsive decision-making, most notably the evidence-based decision to replace the planned choropleth map with a more universally accessible combination of line and bar charts. Key risks have been formally assessed using a probability and impact matrix and mitigated where feasible within the constraints of a proof-of-concept project. The product meets the core functional and non-functional requirements established in assignment one and provides a strong foundation for a production-ready public sector data communication tool.

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